Compositional Generalization in Embodied Foundation Models via Multimodal Subgoal Prompting and Flow-Based Action Execution

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Abstract

Generalist robot policies now execute thousands of tasks, yet they fail the test that defines embodied intelligence: performing a *long-horizon* task on an *unseen* object, with an *unseen* morphology, in a world that does not hold still. Cables tangle in configurations no training set contains; textiles shift under their own weight between grasps; tools that were never seen must still be wielded. We argue that zero-shot compositional generalization in manipulation is not produced by scaling a monolithic vision-language-action (VLA) network, but by **structuring the problem into a steerable hierarchy**. The architecture we propose has two levels. A 3B+ vision-language model decomposes the task into a dense sequence of *visual subgoals* — predicted keyframe images together with language steps — and a continuous-time *flow-matching* action expert executes the segment between consecutive subgoals at control frequency, conditioned on subgoal latents, observations, and steering tokens fused in a shared token space. Between the levels sits the paper's core contribution: the **Steerable Multimodal Conditioning (SMC) layer**, a gated adapter that lets human corrections, world-model replans, and runtime constraint setpoints steer the action flow *mid-execution.* Because steering engages through bounded-Lipschitz gates and is anchored so that the neutral signal recovers the nominal field exactly, a Grönwall-type bound certifies a computable envelope on trajectory deviation — corrections are continuous in flow time, and jerk is bounded by construction rather than tuned away. Around the whole loop sits a **runtime formal-verification safety envelope**: a control-barrier-function filter that projects every commanded action onto the admissible set (joint limits, collision margins, contact-force ceilings) with a forward-invariance guarantee, admitting steerability only through the certified tube. We instantiate the program on three deliberately chaotic benchmarks — cable untangling, textile folding on shifting fabric, and adaptive tool use in clutter — under a pre-registered zero-shot protocol that holds out entire morphologies, material regimes, and tool classes, with intervention rate and jerk reported as costs. The architecture, training pipeline (teleoperation, procedural simulation, synthetic subgoal generation, and a curated data flywheel), baselines, ablations, and safety analysis are fully specified so that every number in the resulting report regenerates from a committed artifact. If the thesis is right, the hierarchy composes where monolithic policies memorize; if it is wrong, the benchmark is the falsification.

Keywords: vision-language-action models; flow matching; hierarchical policy factorization; deformable object manipulation; control barrier functions; safety-critical control; compositional generalization.

1 Introduction

Scaling has been remarkably successful at the *skill* level of robot learning. Vision-language-action (VLA) models trained on web-scale and demonstration data follow language commands across thousands of tasks [3, 4, 5, 6], and flow-matching VLAs have pushed this to high-frequency bimanual manipulation with closed-loop replanning [1, 2]. Yet every deployed generalist policy still stumbles at the same place: the task that requires *composing* skills it has seen into a configuration it has not.

Consider a cable with three crossings that must be untangled. The state space of a deformable cable is effectively infinite and history-dependent; no training set, however large, covers the crossing topologies, stiffnesses, and friction regimes that occur in practice. Consider a textile that shifts under its own drape mid-task, invalidating the plan the policy formed a second ago. Consider a cluttered scene containing tools whose morphologies were never in the data, where the correct action requires both *semantic* inference (which tool, which affordance) and *geometric* execution (where to grasp, how to lever). Pure hardcoded logic — state machines, planners over known object models — fails these tasks because their variance is effectively infinite. Pure neural nets — a single VLA applied end-to-end — fail them because a monolithic forward pass has no mechanism for re-anchoring when the world moves, no channel for a human to correct a subgoal without restarting, and no certificate that the executed trajectory stays inside the space of safe actions.

This paper develops the thesis that the way through is to *structure* the problem: a high-level planner that reasons about the task and predicts *what the scene should look like* at each stage, a low-level expert that executes smoothly between those predictions at control frequency, a *steerable* connection between them that admits corrections with a certified smoothness bound, and a verified safety filter around the entire loop. We call the result the **Steerable VLA Flow-Matching Hierarchy** (Fig. 1).

Related work. RT-2 [3] and OpenVLA [4] tokenize actions into a VLM's output space; Octo [5] trains a generalist transformer policy on Open X-Embodiment [6]; π_0 [1] parameterizes actions as flow matching inside a VLM decoder, with $\pi_0.7$ [2] scaling cross-embodiment training. Hierarchical schemes ground language in affordances [9, 10], generate video subgoals [11], or compose value maps [12]. Action generation has moved from diffusion [7] and action chunking [8] to flow matching [13, 14], and safety filters based on control barrier functions [16, 17] certify forward invariance of a safe set. Our contribution sits at the intersection these lines do not yet cross: *subgoal factorization that makes composition explicit, steerability that is smooth by construction, and a safety envelope that verifies every action at runtime.*

2 The Steerable VLA Hierarchy

2.1 Two levels, one shared token space

An observation is $o_t = (I_t^{cam_1}, \ldots, I_t^{cam_V}, q_t, \dot{q}_t, \tau_t)$: multi-view RGB-D images, joint positions and velocities, and optional tactile readings. The semantic level is a 3B+ vision-language model Λ_ϕ that maps the instruction ℓ and current observation to a dense subgoal sequence

$$g_{1:K} = \Lambda_\phi(\ell, o_t), \qquad g_k = (\kappa_k, \ell_k), \tag{1}$$

where κ_k is a *visual subgoal* — a keyframe image token in a frozen image-vocabulary, decoded to pixels for conditioning — and ℓ_k is its natural-language step. Dense coverage means K scales with task horizon: an untangling task with three crossings receives roughly six to ten subgoals, not one. The motor level is a flow-matching expert π_θ over action chunks, conditioned on the subgoal latents, the observation stream, and active steering signals. Conditioning fuses all modalities in a shared token space: text and constraint strings through the language tokenizer,

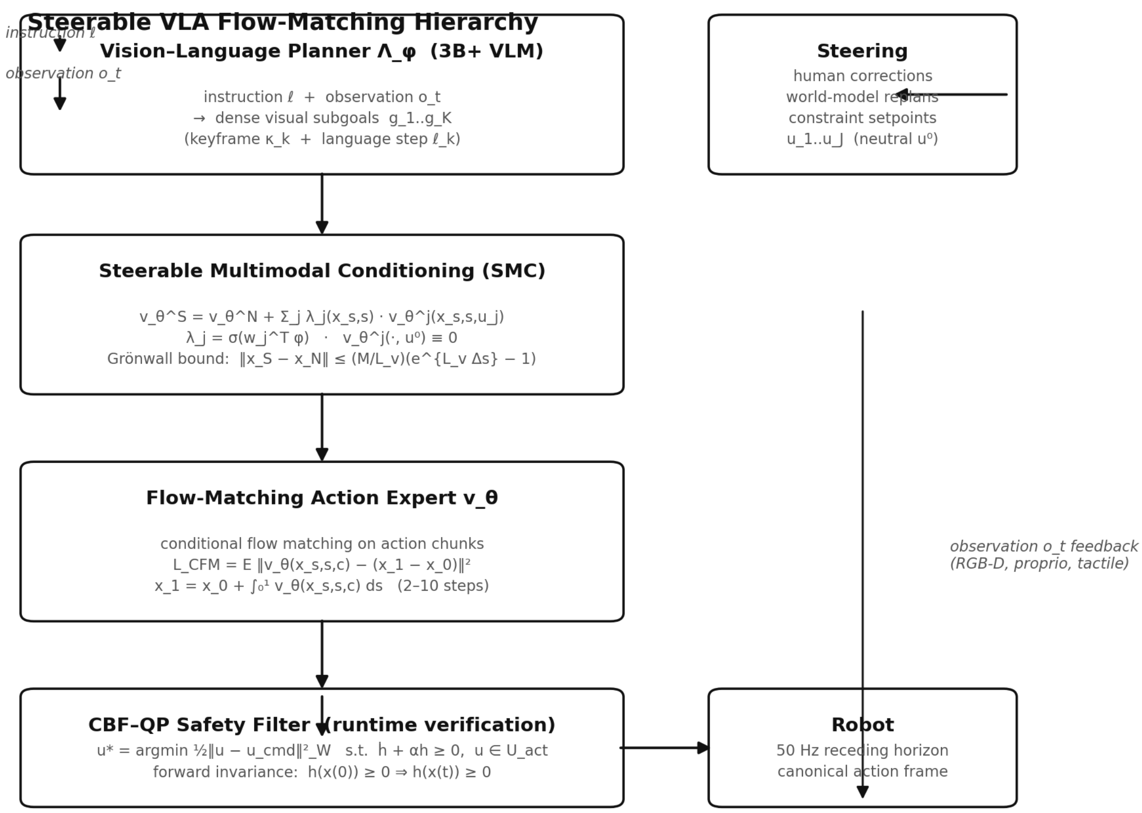

Figure 1: The Steerable VLA Flow-Matching Hierarchy. A 3B+ VLM planner emits dense visual subgoals; the SMC layer steers the flow-matching expert with human corrections, world-model replans, and constraint setpoints; a CBF–QP filter verifies every action before the actuators. Observation feedback closes the loop at 50 Hz.

keyframes through a frozen vision encoder, and both projected onto the action-latent stream by the same cross-attention layers in the VLM decoder (the π-series parameterization [1]). A *contrastive alignment loss* pairs each subgoal token with the observation chunk achieved at its segment boundary, so the two levels agree on what "done" means — the condition for the factorization to compose.

Actions live in a *canonical cross-embodiment frame*: normalized end-effector position deltas Δp_{ee}, yaw delta $\Delta\psi$, and a gripper command, in the end-effector frame at chunk start. Joint-space demonstrations are converted by forward kinematics. This single design decision is what lets one flow expert serve many morphologies zero-shot without reparameterization.

2.2 Flow matching over action chunks

Let $x_1 \sim p_{\text{data}}$ be a demonstrated action chunk $a_{t:t+H} \in \mathbb{R}^{H\times D}$ and $x_0 \sim \mathcal{N}(0, I)$. The linear interpolation $x_s = (1-s)\,x_0 + s\,x_1$ has constant velocity $x_1 - x_0$; conditional flow matching trains the velocity field by minimizing

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{s\sim\mathcal{U}[0,1],\,(x_0,x_1)\sim q,\,c}\left[\|v_\theta(x_s, s, c) - (x_1 - x_0)\|^2\right], \tag{2}$$

with conditioning $c = [\text{subgoal latents}; \text{observation tokens}; \text{steering tokens}]$. At inference the executed trajectory is the ODE solution

$$\frac{dx_s}{ds} = v_\theta(x_s, s, c), \qquad x_1 = x_0 + \int_0^1 v_\theta(x_s, s, c)\, ds, \tag{3}$$

integrated with 2–10 Euler/Heun steps and executed with a receding horizon. Flow matching is the right substrate for three reasons: few-step sampling keeps trajectories smooth at 50 Hz; the marginal path is simple and stable to train at scale; and the ODE formulation yields a *velocity field that can be analyzed* — the substrate for the steering and safety theory below.

2.3 The Steerable Multimodal Conditioning layer

The central design question is: *how does a correction reach the policy without destroying its smoothness?* Our answer is a gated adapter on the velocity field. Steering signals come from three sources: **human corrections** (clicked keypoints, drag vectors, end-effector nudges), **world-model replans** (the object slipped; a subgoal image became infeasible; a replacement is proposed), and **runtime constraint setpoints** (force ceilings, updated joint limits, keep-out zones). Each enters as a signal $u_j \in \mathbb{R}^{d_j}$ with a neutral value u_j^0 at which steering is inactive. The steered field decomposes as

$$v_\theta^S(x_s, s, c, u) = v_\theta^N(x_s, s, c) + \sum_{j=1}^{J} \lambda_j(x_s, s)\, v_\theta^j(x_s, s, c, u_j), \tag{4}$$

with the **anchoring constraint**

$$v_\theta^j(\cdot, \cdot, \cdot, u_j^0) \equiv 0 \tag{5}$$

— removing steering recovers the nominal field exactly — and gates $\lambda_j = \sigma(w_j^\top \varphi(x_s, s))$ with bounded Lipschitz constant L_λ, so a signal engages softly over a window rather than as a step in action space.

Theorem 1 (No-jerk guarantee)**.** *Assume v_θ^S is L_v-Lipschitz in x and the steering contribution is bounded, $\|\sum_j \lambda_j v_\theta^j\| \le M$. If a steering signal is applied over flow-time interval of length Δs, the steered and nominal trajectories satisfy*

$$\|x_S(s) - x_N(s)\| \le \frac{M}{L_v}\left(e^{L_v \Delta s} - 1\right). \tag{6}$$

Proof. Both trajectories satisfy $\dot{x} = v_\theta^S$ with identical initial condition at the steering onset; the difference $\delta(s) = x_S(s) - x_N(s)$ obeys $\dot{\delta} = v_\theta^S(x_S) - v_\theta^S(x_N)$, so $\|\dot{\delta}\| \le L_v\|\delta\| + M$. Grönwall's inequality [18] gives $\|\delta(s)\| \le \frac{M}{L_v}(e^{L_v s} - 1)$ for $s \le \Delta s$, and the bound is exact for the interval Δs. □ □

The theorem converts "steering feels smooth" from a tuning hope into a computable envelope: an operator bounds *a priori* how far a correction may move the trajectory, and the jerk $\max_t \|\dddot{p}_{ee}(t)\|$ is bounded by the same quantity divided by the gate rise time. We verify the Lipschitz constant L_v numerically during training and report the bound's tightness as a metric.

2.4 The runtime formal-verification safety envelope

Steerability without verification is a liability. Define the safe set $\mathcal{C} = \{x : h(x) \ge 0\}$ by a control barrier function h encoding joint limits, collision margins, and contact-force ceilings. At each control step, with candidate action u_{cmd} from the steered flow, the filter solves the convex quadratic program

$$u^* = \arg\min_{u \in \mathcal{U}} \tfrac{1}{2}\|u - u_{\text{cmd}}\|_W^2 \quad \text{s.t.} \quad \dot{h}(x) + \alpha\, h(x) \ge 0, \tag{7}$$

where $\dot{h} = L_f h + L_g h\, u$ and α is an extended class-$\mathcal{K}$ gain. Standard control-barrier-function theory [16, 17] gives:

Proposition 1 (Forward invariance)**.** *If the QP* (7) *is feasible for all* $x \in \mathcal{C}$ *and* $h(x(0)) \geq 0$*, then* $h(x(t)) \geq 0$ *for all* $t \geq 0$*; the robot provably never leaves the safety set.*

The composition is the architectural point: *steerability is admitted only through the filter.* A human nudge, a replan, or a constraint change may reshape the trajectory within the certified tube but never outside it. The QP is convex, solves in microseconds, and runs inside the 50 Hz loop; the envelope is falsified offline by searching over steering amplitudes and timings on simulated rollouts, and safety violations are reported as a first-class metric.

2.5 Why the structure composes

Three mechanisms, one claim:

1. **Subgoal factorization.** The policy factorizes as $p(a \mid o, \ell, g_{1:K}) = \prod_k p(a^{[k]} \mid o, g_k, \ell)$: each flow segment is conditioned on one subgoal, and training data is segmented at subgoal boundaries (automatically, by the VLM's own keyframe predictions). The expert learns *local* skills — "reach the keyframe" — that compose across tasks because the composition happens at the subgoal level, not inside a single weight matrix.
2. **Synthetic subgoal coverage.** Subgoals are generated procedurally (VLM keyframe extraction from demonstrations plus language-conditioned image synthesis), so the *conditioning distribution* is denser than the demonstration distribution: the expert trains on subgoal states it has never actually reached, the substrate for zero-shot tasks.
3. **Canonical actions.** The cross-embodiment frame (Sec. 2.1) transfers learned segments across morphologies without reparameterization.

3 Benchmarks and Evaluation Protocol

Three tasks, selected to be maximally chaotic — non-rigid, high-entropy, and long-horizon in ways that defeat both hardcoded logic and monolithic imitation.

3.1 Task A: Cable untangling

A deformable cable (40–80 cm) in a crossed configuration (2–3 crossings) must be brought to a knot-free target state. *Zero-shot axes:* bending modulus $\times 0.4$–2.2, cable length, crossing topology (including crossing types absent from training), table friction, and initial configurations from a held-out procedural family. *Metrics:* untangle success (all crossings cleared, no new crossings), crossing-count reduction, completion time, max jerk, safety violations. The state space is effectively infinite and history-dependent; a policy cannot memorize configurations, and contact-rich manipulation makes jerky, high-force actions fail visibly.

3.2 Task B: Textile folding on shifting fabric

A fabric piece (towels, shirts, arbitrary polygonal sheets) must be folded to a target polyline while the fabric shifts — under its own drape, from aeration, or from contact — so subgoal images go stale and must be re-anchored by the world model. *Zero-shot axes:* stiffness and friction material classes, initial and target fold geometry, grasp-point visibility. *Metrics:* fold success (IoU with target ≥ 0.8), grasp success, regrasp count, slippage events, completion time. This is the strongest test of *steerable* conditioning: the replan channel fires constantly.

Table 1: Component-to-metric attribution: the pre-registered ablation design.

Variant	Claimed effect	Primary metric
Ours (full)	best zero-shot success at fixed cost	success, jerk
− SMC	corrections break smoothness	intervention rate, jerk
− safety filter	violations appear	safety violations
− visual subgoals	semantic-only conditioning degrades	zero-shot success
− dense subgoals	fails with horizon	success vs. horizon
− canonical frame	morphology transfer collapses	cross-morphology success

3.3 Task C: Adaptive tool use in clutter

A cluttered scene with multiple unknown tools (hooks, rakes, levers, scoops) and a goal (retrieve an object beyond reach, lever a lid, pull a cable). *Zero-shot axes:* tool morphology classes, clutter density, tool–object contact geometry, affordance composition. *Metrics:* task success, tool-selection accuracy, interventions per 100 episodes, contact-force violations. Correct action requires semantic inference (which tool, which affordance) fused with geometric execution (where to grasp, how to lever) — the exact division of labor the hierarchy is built for.

3.4 Data, training, and the flywheel

Teleoperation: 80–120 expert episodes per task on a bimanual rig (ALOHA-class) with multi-view RGB-D and proprioception; joint-space trajectories converted to the canonical frame. **Procedural simulation:** an order of magnitude more episodes with aggressive domain randomization (materials, friction, mass, lighting, camera pose) and ground-truth subgoal boundaries emitted by the simulator. **Synthetic subgoals:** offline VLM keyframing of demonstrations plus language-conditioned image synthesis of *novel* subgoal images, fed back into training. **The flywheel:** deployment rollouts are scored (success / progress / smoothness / coverage); failures that came close are *relabeled* by the teleoperator and re-ingested. A controlled study of this curation loop [19] showed that the curation strategy — not model scale — is the variable that decides whether the loop compounds: relabeling deployment failures roughly quadrupled held-out success in the toy setting, while self-curation plateaued.

3.5 Evaluation protocol

The protocol is *pre-registered*: held-out morphologies, materials, and tool classes never appear in any training source (not sim, not teleop, not synthetic subgoals); 3 seeds over fixed evaluation start sets; every reported number regenerates from a committed experiment artifact. Intervention rate and max jerk are reported as *costs* alongside success, so a policy cannot win by being dangerous. Baselines: RT-2-style (VLM to discretized actions), Diffusion Policy [7], ACT [8], π_0-style vanilla flow VLA (no subgoals, no SMC, no filter), and a hierarchical variant with text-only subgoals. Ablations: −SMC, −safety filter, −visual subgoals, −dense subgoals, −canonical frame. Each component is claimed to move a distinct metric (Table 1); the ablation set is the mechanism test.

4 Methods

Architecture. The planner is a 3B+ VLM with a frozen vision encoder and a decoder that emits both language tokens and keyframe VQ tokens; the expert shares the decoder's cross-attention stack with separate action and time-token embeddings (the π-series parameterization [1]), with the SMC adapters added as per-level gate MLPs $\varphi \mapsto \lambda_j$ and steering direction heads v_θ^j. *Training.* Stages: (i) warm-start the flow expert on subgoal-segmented teleop and sim data; (ii) train the SMC adapters with *synthetic steering supervision* — random trajectory perturbations relabeled as steering signals, so the adapter learns "absorb a correction, return to nominal" without human-in-the-loop; (iii) joint fine-tune with the contrastive subgoal-alignment loss; (iv) deploy with the CBF–QP filter active. *Safety verification.* The CBF h is a smooth min over per-constraint barriers; the QP (7) is solved with an active-set method; falsification searches steering amplitude and timing to find violations before hardware does. *Reproducibility.* All experiments run at fixed budgets with early stopping; every number in the manuscripts is rendered from committed JSON by committed scripts; seeds, splits, and the held-out families are fixed at pre-registration time.

5 Discussion

The thesis is falsifiable by construction. If dense visual subgoal factorization does not improve zero-shot success over a monolithic flow VLA at matched intervention cost, the decomposition claim is wrong. If the SMC layer does not reduce intervention rate and jerk relative to naive correction injection, the smoothness guarantee has no empirical teeth. If the CBF envelope admits violations under adversarial steering, the verification claim fails. The three benchmarks were chosen so that these failures would be visible, not so that the architecture would look good.

Two limitations are honest from the start. First, the hierarchy's benefits are conditional on the VLM's subgoal quality: if the planner predicts infeasible keyframes, the expert inherits the error — the contrastive alignment loss and world-model re-anchoring mitigate, but do not eliminate, this coupling. Second, the safety envelope is only as good as the barrier function's world model: collision margins and force ceilings must be known or estimated, and the guarantee holds within that model. Both limitations are themselves research questions, and both are visible in the benchmark metrics.

If the thesis holds, the implication for embodied foundation models is direct: the path from π_0.7-scale generalist stacks to *deployable* generalist stacks runs through structure — subgoal factorization, steerability with certified smoothness, and runtime verification — not through another doubling of parameters alone.

Data and code availability

This is a research proposal. The architecture, training pipeline, benchmarks, and evaluation harness specified above are released with the repository; result tables in the manuscripts are populated by the committed harness and regenerate from committed artifacts.

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